



Application Services Assignment

Intro

This assignment is intended to evaluate your hands-on capabilities, give you a taste for the work we do, and show us how you communicate complex subjects in writing. You can refer to:

[Application Services](#), [Cloudflare Docs](#) and the [Cloudflare Dashboard](#).

Please keep in mind that, while we can help point you in the right direction, the objective is to evaluate your ability to learn new concepts quickly through effective research and self-discovery. The Cloudflare offering is vast, and it is crucial that you are comfortable researching answers on topics you may not be familiar with. This assignment is designed to be completed under free tier services.

Deliverables

1. A working application with instructions needed to access any of the above.
2. A written report that addresses the following:
 - a. Describe the steps followed to implement the technical requirements, including screenshots showing the configuration and testing evidence.
 - b. What are the relevant use cases the products are being used for?
 - c. How did you fill in the gaps (if any) in your knowledge during the process?
 - d. How do you imagine that a target customer will find this experience?

Pre-requisites

1. Create a Cloudflare Free plan account or use an existing account.
2. Please add a domain (new or existing, e.g. yourwebsite.com) to Cloudflare.
3. Activate on Cloudflare by following the steps to change the nameservers at your registrar.

Steps

1. Create an origin web server on a platform of your choosing. This could be in AWS, Google Cloud, DigitalOcean, your Raspberry Pi, etc. This web server must run an endpoint that returns all HTTP request headers in the body of the HTTP response. The

web server can be something that you have written yourself (e.g. in JavaScript, Python, etc) or by using a 3rd party application.

2. Proxy traffic to this server through Cloudflare. Add necessary configurations on Cloudflare.
3. Secure the communication between Cloudflare and your Origin Server with a non-Cloudflare provisioned TLS certificate using at least the Full-Strict mode on Cloudflare.
4. Install and configure Cloudflare Tunnel on your origin server using a subdomain called “tunnel”, e.g. tunnel.yourwebsite.com. Make connections proxied to your server protected using this tunnel.
5. Configure an SSO IdP (Identity Provider) of your choosing within Cloudflare Zero Trust.
6. Lock down access for a particular path for your Cloudflare Tunnel subdomain (e.g. tunnel.yourwebsite.com/secure) and only allow access for yourself (with the previously configured IdP of your choosing) and users with an @cloudflare.com email address. Ensure nobody can bypass Cloudflare and access your server’s IP directly.
7. Create a Cloudflare Worker.
 - a. Serve a Worker on the tunnel.yourwebsite.com/secure path that returns identity information for the authenticated user.
 - b. The HTTP response body should contain: “\${EMAIL} authenticated at \${TIMESTAMP} from \${COUNTRY}”
 - c. \${COUNTRY} should be an HTML link that when clicked navigates to tunnel.yourwebsite.com/secure/\${COUNTRY} and displays the appropriate country flag.
 - d. The flag asset should be stored in a private R2 bucket.
 - i. Create this Worker using the Wrangler CLI, upload your Workers code to a public Git repository for your implementation.
 - ii. The /secure response should be returned as HTML
 - iii. The /secure/\${COUNTRY} response should use an appropriate content type